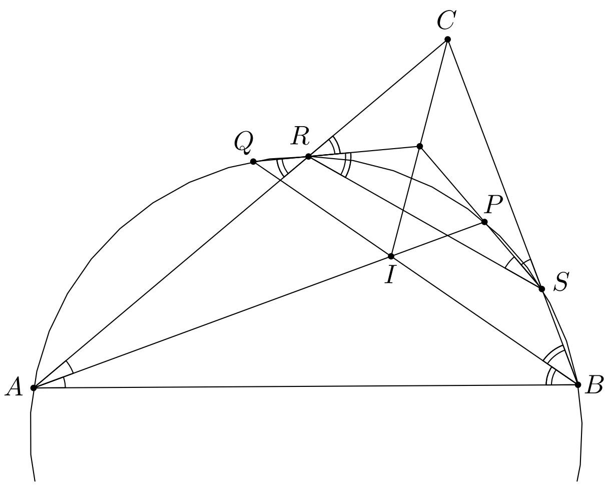
Problem 93. Let I be the incentre of $\triangle ABC$ and let a circle Γ through the points A and B intersect the lines

- AI at points A and P,
- BI at points B and Q,
- AC at points A and R, and
- BC at points B and S,

with none of the points A, B, P, Q, R and S coinciding and the points R and S being interior points of the line segments AC and BC, respectively.

Prove that the lines PS, QR and CI are concurrent.

Solution.



Since points A, B, R and S lie on a common circle, we have $\angle BSR = 180° - \angle RAB = 180° - \angle CAB$, and therefore $\angle RSC = \angle CAB$. Similarly, $\angle CRS = \angle ABC$ also holds.

If P lies in the interior of $\triangle ABC$, we have $\angle RSP = \angle RAP = \frac{1}{2}\angle CAB$. This means that PS bisects the $\angle CSR$.

If Q is outside of $\triangle ABC$, we have $\angle QRA = \angle QBA = \frac{1}{2}\angle ABC$, and in this case QR also bisects $\angle SRC$.

Independent of the positioning of P and Q with respect to the triangle, we therefore see that QR, PS and CI are the bisectors of the interior angles of $\triangle CRS$, and therefore, are concurrent at its incentre. □

Problem 94. On a circle 2020 points are marked. Each of these points is labelled with an integer. Let each number be larger than the sum of the preceding two numbers in clockwise order.

Determine the maximal number of positive integers that can occur in such a configuration of 2020 integers.

Solution. The answer is 1009.

Let the points be labelled as $a_0, a_1, \ldots, a_{2019}$ clockwise with cyclical notation, i.e. $a_{k+2020} = a_k$ for all integers k.

Lemma 1. In a valid configuration, no two neighbouring numbers can be both non-negative.

Proof. Assume that there exist neighbouring numbers a_{k-1} and a_k which are both non-negative. We get $a_{k+1} > a_k + a_{k-1} \geqslant a_k$, with the first inequality following from the problem statement and the second from $a_{k-1} \geqslant 0$. Since now also a_k and a_{k+1} are both non-negative, we analogously get $a_{k+2} > a_{k+1}$, and then $a_{k+3} > a_{k+2}$, and so on, until we have

$$a_{k+2020} > a_{k+2019} > \cdots > a_{k+1} > a_k = a_{k+2020},$$

a contradiction.

Hence at most every second number can be non-negative. Next we will show that these are still too many non-negative numbers.

Lemma 2. In a valid configuration, it is not possible that every second number is non-negative.

Proof. Assume that this is the case, so without loss of generality, we assume that $a_{2k} \geqslant 0$ and $a_{2k+1} < 0$ for all integers k. Then we get $a_3 > a_2 + a_1 \geqslant a_1$, where again the first inequality follows from the problem statement and the second from $a_2 \geqslant 0$. Analogously, we get $a_5 > a_3$, and then $a_7 > a_5$, and so on until we have

$$a_1 = a_{2021} > a_{2019} > a_{2017} > \cdots > a_3 > a_1,$$

a contradiction.

Therefore a configuration with more than 1010 non-negative numbers is not possible because otherwise by the pigeonhole principle we would have two neighbouring non-negative numbers, which is not allowed according to the first lemma. A configuration with exactly 1010 non-negative numbers contradicts either the first or the second lemma.

With 1009 positive numbers and 1011 negative numbers, we have the following example of a configuration satisfying the problem statement:

$$-4039, 1, -4037, 1, -4035, 1, -4033, \ldots, 1, -2023, 1, -2021, -2019. \qquad \square$$

Problem 95. Determine all digits d such that for each integer $k \geqslant 1$ there exists an integer $n \geqslant 1$ with the property that the decimal representation of n^9 ends with at least k digits d.

Solution. For $d = 0$ we easily find 10^ℓ with any sufficiently large integer ℓ such that $9\ell \geqslant k$. For $d \in \{2, 4, 6, 8\}$ the number n^9 is even and therefore also n must be even, and hence n^9 must be divisible by 2^9. However, numbers ending with $222, 444$ or 666 are already not divisible by 8, and numbers ending with 8888 are not divisible by 16. Therefore, there does not exist a solution for these values of d.

Similarly, for $d = 5$ the number n^9 is divisible by 5, therefore also n itself is divisible by 5 and therefore, n^9 must even be divisible by 5^9. However, numbers ending with 55 are not divisible by 25.

For $d \in \{1, 3, 7, 9\}$, let $b = \underbrace{\overline{ddd\ldots d}}_{k}$ (that is, b is the integer with k digits d). Since

$$\gcd(9, \varphi(10^k)) = \gcd(9, 4 \cdot 10^{k-1}) = 1,$$

by the Euclidean algorithm, there exist integers x and y such that $9x + \varphi(10^k)y = 1$. We claim that $n = b^x$ has the desired property. Because of $\gcd(b, 10^k) = 1$ this can be easily demonstrated using Euler-Fermat:

$$(b^x)^9 = b^{9x} \equiv b^{9x+\varphi(10^k)y} = b^1 = b \pmod{10^k}. \qquad \square$$

Problem 96. Consider a board consisting of $n \times n$ unit squares where $n \geqslant 2$. Two cells are called neighbours if they share a horizontal or vertical border. In the beginning, all cells together contain k tokens. Each cell may contain one or several tokens or none.

In each turn, choose one of the cells that contains at least one token for each of its neighbours and move one of those to each of its neighbours. The game ends if no such cell exists.

(a) Find the minimal k such that the game does not end for any starting configuration and choice of cells during the game.

(b) Find the maximal k such that the game ends for any starting configuration and choice of cells during the game.

Solution. The answer for (a) is $k = 3n^2 - 4n + 1$ and for (b) is $k = 2n^2 - 2n - 1$

(a) If each cell contains one token less than the number of its neighbours, the game cannot even start. On the other hand, if there is one token more then by the pigeonhole principle there will always exist at least one cell with sufficient tokens to make the next move.

Therefore, the desired quantity is the sum of all numbers of neighbours minus the number of all cells plus 1. If one adds $4n$ cells around the n^2 given cells, each original cell has four neighbours and each new cell has contributed one neighbour. We get $k = (4n^2 - 4n) - n^2 + 1 = 3n^2 - 4n + 1$.

(b) It is easy to see that an unlimited number of turns must eventually have brought tokens to all cells and that also every pair of neighbours must have exchanged tokens because otherwise tokens would accumulate in unlimited number in one of the inactive cells.

But each time, a neighbouring pair first exchanges tokens, we can reserve this first token to stay always between these two neighbours. Therefore, the game will certainly end if there are less tokens then neighbouring pairs.

Conversely, if the number of tokens equals the number of neighbouring pairs, we can find a neverending game in the following way: Colour the cells black and white in a checkerboard fashion and assign to each black cell a number of tokens that equals the number of its neighbours. Now we will simply choose all the black cells until all tokens are on the white cells, then repeat with the white cells and then iterate from the start.

The desired quantity is therefore the number of neighbouring pairs minus 1. Since the number of neighbouring pairs is half of the first expression in the computation of part (a), we get $k = 2n^2 - 2n - 1$. $\square$